Highlights

Context-Aware Interval Transcription for Visual Speech Recognition

Author information to be inserted

- Context admission and requested-output ownership are formulated as separate decisions.
- Prediction-aligned projection uses a frozen VSR model and native visual CTC scores.
- Matched decoding separates contextual encoding gains from transcript readout gains.
- Context availability and query reuse determine the accuracy–workload trade-off.

Context-Aware Interval Transcription for Visual Speech Recognition

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Abstract

Transcribing a selected video passage requires controlling the returned text while retaining useful neighboring visual evidence. We formulate context-aware interval transcription by separating the requested text interval from the permitted observation interval. A training-free implementation, prediction-aligned projection, recognizes the available context, aligns its predicted tokens using native visual CTC scores, and returns complete words assigned to the request. We evaluate this procedure together with crop-first recognition, contextual direct decoding, identical-transcript allocation controls, and reusable full-clip lookup. Across 14,650 clips from four corpora in three languages, development-selected projection reduces requested-interval error by 2.37–8.17 percentage points relative to crop-first recognition. Matched-decoder comparisons distinguish the benefit of contextual encoding from the additional, decoder-dependent effect of transcript projection. Real neighboring video also outperforms equal-length repeated-boundary padding. A 4,006-clip duration study relates these effects to the context actually available around a request: one-second LRS2 targets improve from 47.48% to 29.98% WER with up to 0.64

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seconds per side. Repeated-query measurements identify a complementary workload distinction: bounded recognition can reduce first-query work, while full-clip transcription benefits from reuse and remains stronger for complete transcripts. The resulting framework connects recognition context, output selection, and computational reuse for passage-level visual transcription.
Keywords: visual speech recognition, interval transcription, visual context, CTC alignment, inference, multilingual evaluation

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1. Introduction

Visual speech recognition (VSR) recovers spoken language from facial motion. Recent systems learn strong representations from large audiovisual collections and decode complete utterances with CTC, attention, or combinations of the two (Graves et al., 2006; Watanabe et al., 2017; Ma et al., 2023). Complete-utterance recognition is an important benchmark, but an interface need not request an entire utterance. A video editor may select a short passage for transcription, or a viewer may request the speech in a particular time interval. When audio is unavailable or unsuitable for the task, a visual recognizer must decide both what context to observe and which recognized words to return.

The obvious solution is to crop the requested interval and run an existing recognizer. This makes the input boundary coincide with the requested output boundary. The coincidence is convenient computationally, but may discard useful speech immediately outside the selection. Neighboring video can help resolve the target's visual ambiguity or preserve temporal structure near the crop boundary. However, passing a larger clip to the recognizer

introduces another problem: the resulting transcript includes speech that was not requested. Context admission and output selection therefore need to be considered together.

We formulate *context-aware interval transcription*: given a requested interval and an allowance of neighboring video, return the transcript assigned to the request. The formulation separates two decisions: which frames support recognition, and where to enforce the output boundary. Contextual direct decoding restricts contextual encoder states before decoding. Our *prediction-aligned projection* instead decodes the allowed video, aligns its predicted tokens with native visual CTC scores, and selects complete words after recognition. Both procedures use the same contextual input and frozen recognizer. Their comparison reveals how the stage of output restriction interacts with decoding.

Three research questions organize the study. First, does genuine neighboring video improve recognition of a fixed target beyond the effect of input length? Second, when does selecting from a contextual transcript help beyond selecting contextual encoder states? Third, how do the context available around a request and the reuse of earlier computation affect the accuracy–cost tradeoff? Crop-first recognition, matched decoders, identical-transcript allocation controls, and full-clip timestamp lookup provide the corresponding comparisons. The full-clip alternative is particularly relevant when an application can process the entire video or already holds a transcript index.

The study contributes an interval-transcription formulation and a concrete prediction-aligned implementation that adapts pretrained VSR to selected-passage requests. A controlled four-corpus evaluation then separates con-

textual encoding, decoder behavior, and temporal output selection. Finally, context interventions, duration grids, and repeated-query measurements establish an empirical account of when bounded recognition is useful and when reusable full-clip processing is preferable. Together, these contributions support passage-level visual transcription through an explicit treatment of context, output scope, and work per request.

2. Related work

Visual context and recognition units.. Visual context is already important in lipreading. Stafylakis et al. (2018) study target-word boundaries and context for word recognition. Sentence recognizers also learn temporal dependencies over their input, while large-scale automatic labeling helps expand their training material (Ma et al., 2023; Yeo et al., 2024). More recently, GLip learns to combine local and global visual features, including contextual enhancement, to improve recognition under visual degradation (Wang et al., 2025). Such learned feature integration addresses representation robustness; our study holds the recognizer fixed and varies the observed interval and returned text scope. Our question concerns inference with these existing recognizers: how can the input retain surrounding speech while the returned transcript is restricted to an arbitrary requested interval? The target is a variable-length unknown text, rather than a class from a fixed word vocabulary.

Visual alignment and keyword localization.. Kim et al. (2023) learn to align a supplied transcript with talking-face video and examine transcript–video mismatches. CTC also supplies a monotonic alignment structure, and CTC segmentation has been used to associate text with long speech recordings

(Graves et al., 2006; Kürzinger et al., 2020). Recognition followed by alignment is also established in speech: WhisperX combines transcription and forced phoneme alignment to obtain word timestamps (Bain et al., 2023). We use this general pattern within a frozen visual recognizer, aligning its own predicted tokens with its native CTC scores. The contribution lies in evaluating how contextual observation and temporal output selection interact for interval requests, with separate controls for context, decoding, and reusable full-clip transcription.

Visual keyword spotting takes a complementary input–output direction. For example, Transpotter receives a keyword representation and video, then localizes the keyword if present (Prajwal et al., 2021). Interval transcription receives a temporal location and predicts its unknown text. Neither a keyword nor the reference transcript is supplied to our inference procedure.

Decoder and corpus conditions.. Hybrid CTC/attention decoding combines monotonic alignment with sequence-level prediction (Watanabe et al., 2017). Comparisons of VSR decoder families show that decoder behavior depends on data and training conditions (Gimeno-Gómez and Martínez-Hinarejos, 2024). Spanish VSR experiments further illustrate the importance of corpus conditions and transfer (Gimeno-Gómez and Martínez-Hinarejos, 2025). Consequently, we report both independently selected systems and matched-decoder controls. Multilingual corpora such as MuAViC enable evaluation beyond English (Anwar et al., 2023), but differences between languages remain entangled with corpus composition and checkpoint training. Our cross-corpus results are replications of an inference comparison, not estimates of a language effect. Recent evaluations further show sensitivity to evaluation distributions

and lexical bias beyond the standard benchmark (Lin et al., 2026), while isolated-word experiments identify differences between model and human recognition patterns (Jain and Harte, 2026). These findings motivate examining the information available around a request and the decoder used to interpret it. Our paired context interventions address that inference question without treating lower transcription error as evidence of a particular human-like perceptual mechanism.

Efficient and selective inference.. Efficiency work can reduce the representation or computation used by a speech model; selective inference instead changes the portion of input needed to answer a request. We examine this second choice without retraining the backbone. Its appropriate alternative is a reusable full-clip transcript with timestamps. The measured workload therefore includes both a first interval request and multiple requests, rather than assuming that a bounded input is always faster.

3. Context-aware interval transcription

3.1. Requested and observed intervals

Let $X = (x_0, \ldots, x_{T-1})$ be a prepared mouth-region video, and let $I = [a, b)$ be the requested interval in frame coordinates. A context allowance (c_ℓ, c_r) defines the permitted input

$$J = [\max(0, a - c_\ell), \min(T, b + c_r)). \tag{1}$$

The recognizer may observe X_J, while the returned text should belong to I. Context is real video from the same clip; no frames are fetched beyond its

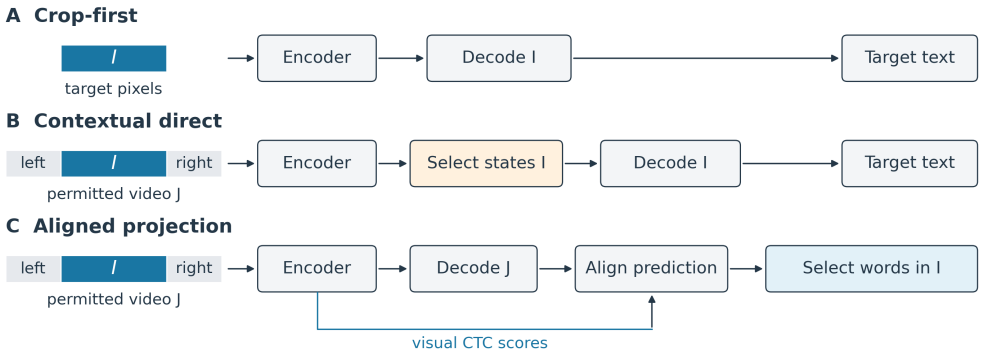


Figure 1: Three ways to transcribe the same requested interval I. Crop-first restricts the video before encoding. Contextual direct encodes the permitted context J and selects target states before decoding. Aligned projection decodes J, aligns its own prediction with visual CTC scores, and selects words assigned to I. B and C observe the same video; all alternatives reuse the same frozen recognizer within a corpus.

boundaries. Thus a nominal context budget is an upper bound, and $J = I$ is possible for short clips.

Write E for the visual encoder and D for a decoder. The encoder also provides visual CTC token scores P_J. The same pretrained parameters are used for all inference alternatives within a corpus/model setting. The experiment distinguishes three operations (Figure 1):

$$\widehat{Y}_A = D(E(X_I)), \tag{2}$$

$$\widehat{Y}_B = D\Big([E(X_J)]_I\Big), \tag{3}$$

$$\widehat{Y}_C = \Pi_I\Big(D(E(X_J)), P_J\Big). \tag{4}$$

Crop-first A restricts pixels before encoding. Contextual direct B encodes J, then restricts the decoder to contextual states associated with I. Projection C decodes all of J, then restricts the returned words. Equations (3) and (4)

use the same permitted visual information. They differ in where the output restriction is applied. The equations motivate an empirical comparison of state-level and transcript-level readout.

3.2. Prediction-aligned projection

The contextual decoder first produces token sequence $\widehat{z}_J$. We then align that sequence monotonically to P_J using a CTC trellis. Alignment is conditioned on predicted tokens, never on the evaluation transcript. The recognizer therefore supplies both the text proposal and the visual scores used to place it.

Tokenizer-aware grouping converts the aligned tokens into complete words. For a predicted word w_k, the union of its constituent token spans gives $[s_k, e_k)$ in the contextual input. Its midpoint is mapped back to the original clip coordinates. Projection returns

$$\widehat{Y}_C = \text{concat}\{w_k : \ a \leq j_0 + (s_k + e_k)/2 < b\}, \tag{5}$$

in predicted order, where j_0 is the start of J. Whole words are returned; the procedure does not cut a subword string at an arbitrary frame. If the requested interval covers the full available input, the native decoded transcript is returned directly. This identity case avoids changing ordinary recognition when there is no outside context.

Projection selects among words in the contextual prediction and preserves their lexical content and order. It cannot replace a misrecognized word with the correct word or recover a word omitted by the contextual decoder. Its role is temporal selection: retaining words assigned to the target and excluding words assigned outside it. Recognition and temporal assignment thus jointly

determine the returned text. The procedure adds alignment and allocation work while retaining the pretrained parameters and native CTC head.

3.3. Controls and unrestricted lookup

Two allocation controls reuse exactly the same decoded sequence as C. Uniform-word allocation places its words evenly across J; uniform-token allocation distributes the predicted tokens evenly and then reconstructs word spans. Both apply the same midpoint selection rule. Their contrast with C isolates the usefulness of predicted visual alignment from the quality of the generated transcript.

Full-clip lookup F recognizes X, constructs predicted word times, and returns words assigned to I. It answers the same target request but has unrestricted visual input. We also score its complete transcript, as a separate endpoint, when examining sliding-window recognition. The two endpoints have different reference denominators and are never treated as interchangeable.

3.4. Decoder comparability

We evaluate greedy CTC, pure CTC prefix beam search, and the recognizer's native hybrid CTC/attention decoding. The pure CTC beam has width 40 and a token-pruning limit of 64; no external language model is added. Selected-system results use the decoder chosen for each arm on development data. Matched greedy and pure CTC results hold that choice fixed across arms.

For hybrid decoding, the output-length cap is derived from the number of observed frames: $|I|$ for A, $|J|$ for B/C, and T for F. In particular, B and C share the same input-derived cap, although B decodes only target states.

Greedy and pure CTC use their complete supplied state sequences rather than an autoregressive cap. Native tokenizer, normalization, and checkpoint interfaces are retained for each model.

3.5. Computation across interval requests

Consider a session with requests $I_1, \ldots, I_q$, and let $\mathcal{J}$ contain their distinct observed intervals. With a fixed decoder, write $r(J)$ for recognition cost and $a(J)$ for the alignment needed to select words. Ignoring the much smaller selection operation, fresh projected and full-clip sessions have the respective cost decompositions

$$\tau_C \simeq \sum_{J \in \mathcal{J}} \Big[r(J) + a(J) \Big], \tag{6}$$

$$\tau_F \simeq r([0, T)) + a([0, T)). \tag{7}$$

Alignment is unnecessary when the request covers the entire decoded input. Exact repeated computations are cached; overlapping but different intervals are not merged into a shared full-clip encoding. The observed interval set, rather than the nominal number of requests alone, therefore determines work. Short clips can make several requested intervals identical. Distinct targets can also yield the same observed interval after clipping context at video boundaries. C can then reuse one contextual prediction, while B may still decode different target-state sequences. Both alternatives reuse an identical contextual encoding when it is available.

This decomposition links computation to both the observed intervals and their reuse. B and C encode the same context, while C decodes a larger state sequence and may require alignment. We measure response cost together with

accuracy, holding the decoder fixed to attribute differences to the inference procedure.

4. Experimental protocol

4.1. Corpora, checkpoints, and evaluation units

The principal assessment contains 11,306 VLR clips, 1,243 LRS2 clips, 934 Spanish clips, and 1,167 ViCocktail clips. VLR and ViCocktail provide two Vietnamese corpus/model conditions; LRS2 provides English; the Spanish evaluation uses the MuAViC/mTEDx material (Nguyen Quoc et al., 2026; Afouras et al., 2022; Anwar et al., 2023; Nguyen et al., 2025). The comparison is paired within each corpus and is not pooled across languages. Recording-source counts are 61, 354, 12, and 20, respectively. These identifiers group clips from the same source and are not verified speaker identities. The prepared clips also differ in duration: means are 5.78 s for VLR, 1.64 s for LRS2, 5.91 s for Spanish and 5.83 s for ViCocktail. Appendix Appendix A reports the distributions, actual context availability and recording-source weights for these exact assessment cohorts.

Figure 2 illustrates the visual input and the distinction between target and context. Figure 3 relates that distinction to the evaluated clip durations. A context allowance specifies a maximum; the actual neighboring evidence depends on clip length and request position.

Each setting uses a frozen, language-compatible checkpoint and its native tokenizer. English uses the published Auto-AVSR visual model trained with LRS2/LRS3/VoxCeleb2/AVSpeech material (Ma et al., 2023). VLR uses our previously adapted T0 checkpoint (seed 17, selected at step 3,000), and



Figure 2: Actual prepared VLR mouth-region inputs for three centered requests. Each row shows nine sampled frames from an observed interval with a two-second target (teal borders) and up to 0.64 seconds of context on each side (orange). Times are relative to the original clip. The inputs retain their original resolution, motion blur and overlays; they are neither retouched nor generated. Examples were selected by clip identifier without using recognition outcomes; distinct recording IDs do not imply distinct speakers. Images adapted from VLR v2 (Nguyen Quoc et al., 2026), CC BY 4.0.

ViCocktail uses our separately trained Vietnamese Auto-AVSR checkpoint (epoch 13). These are local model instances, not checkpoints released by the respective dataset authors. Spanish uses the released `vsr-low` model of Yeo et al. (2024), trained with mTEDx, VoxCeleb2, and AVSpeech material. Its release filename is `es_mted_vox_avs_wer_45.7.pth`; the English release is `vsr_trlrs2lrs3vox2avsp_base.pth`. The output vocabularies contain 2,002 classes for the two Vietnamese models, 5,049 for English, and 1,006 for Spanish, with the corresponding tokenizer and special-token conventions

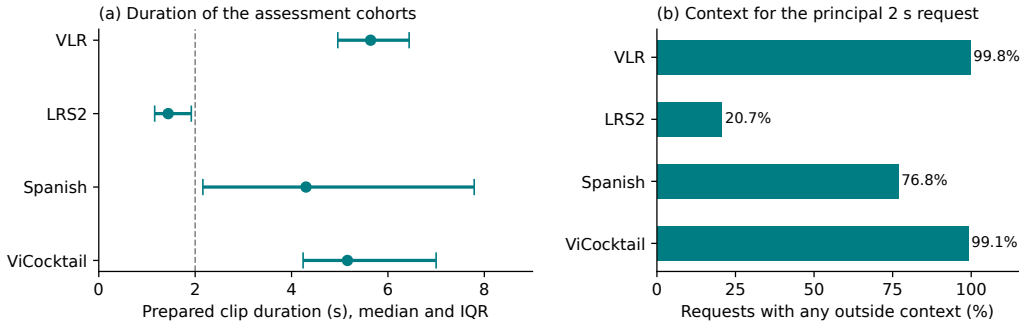


Figure 3: Geometry of the exact four assessment cohorts. (a) Median clip duration with interquartile range; the dashed line marks the nominal two-second target. (b) Fraction of principal requests with any real video outside their target. These are cohort descriptions, not whole-dataset or language-population estimates. The short LRS2 clips provide less opportunity to admit context.

retained. The inference procedure reuses these checkpoints without additional training.

Decoder calibration and assessment are clip- and source-disjoint within the present protocol. Calibration sizes are 512 for VLR, English, and ViCocktail, and 286 for Spanish. Assessment membership is frozen before these comparisons. The material was available during earlier research, so this is an expanded controlled evaluation rather than a newly collected independent replication. Eligibility requires usable video and word timing; one VLR clip and 78 of 1,012 Spanish candidate clips are timing-ineligible. No clip is removed because of its recognition error.

Calibration compares greedy CTC, pure CTC beam search, and hybrid decoding with CTC weights 0.1, 0.5, and 0.9. Beam searches use width 40 and the hybrid length bonus is zero. VLR selects greedy decoding for A/B/C and hybrid weight 0.5 for F. The other three settings select hybrid weight

0.1 for A/C/F and 0.5 for B. Uniform allocation inherits C's prediction and decoder. These choices are retained throughout the assessment and duration experiments; matched-decoder comparisons remove this between-arm choice.

We report corpus-level edit error

$$\mathrm{WER} = 100\,\frac{\sum_i (S_i + D_i + I_i)}{\sum_i N_i}, \tag{8}$$

where the counts refer to the same requested reference text for all compared arms. Unicode NFC normalization, lowercasing, and whitespace normalization are fixed. Vietnamese uses whitespace-delimited orthographic units, which need not coincide with linguistic words. Absolute error rates therefore do not support a cross-language difficulty ranking.

4.2. Target construction and reference timing

The principal setting uses a target of at most 50 frames at 25 fps and up to 16 context frames on either side. Start, center, and end placements are assigned deterministically from clip identifiers. Clips shorter than the target are retained as whole-clip targets, rather than excluded to favor context methods. Within an experimental cell, all arms have the same target reference.

Word midpoints from the fixed reference timing determine target membership. Whole-clip targets retain the complete original transcript. The reference transcript and its timing are accessed only for scoring; inference uses video, predicted tokens, and predicted alignment. VLR retains the supplied ASR-derived transcript and word spans. LRS2 and Spanish use audio-CTC forced alignment of their corpus transcripts; ViCocktail uses its native test labels, reported as manually transcribed by the dataset authors, with audio-CTC word alignment. Audio is used only to prepare evaluation references,

never by the visual recognizer or prediction-alignment procedure. These automatic word times are not independent human boundary annotations. Reference-clock perturbations of $\pm 40, \pm 80, \pm 160$ ms and an overlap-based ownership convention examine sensitivity to this evaluation choice.

4.3. Controlled experiments

The duration experiment varies nominal target length over 1, 2, and 3 seconds and maximum context per side over 0, 0.32, and 0.64 seconds. It includes 1,024 clips each for VLR, LRS2, and ViCocktail, and all 934 eligible Spanish clips. The decoder choices remain those calibrated at the principal operating point; each duration cell is evaluated without selecting a new decoder. Within each target length, it compares the same clips and requested text. Changing target length changes the reference content, so the smallest raw WER across lengths is not interpreted as an optimal segmentation duration.

Context interventions use 256 clips per corpus. They compare real context with equal-length repeated-boundary padding, reversed margins, and left-only or right-only context while preserving target pixels. The primary padding contrast uses clips that actually have outside context. The full-transcript experiment uses 512 clips per corpus. For nominal target budget L, a clip of T frames is partitioned into $K = \lceil T/L \rceil$ balanced, nonoverlapping target intervals. Direct and projected sliding share that schedule and concatenate their outputs in time order. A further control joins contextual target CTC scores before decoding once, separating text concatenation from repeated local recognition.

For query costs, 32 duration-stratified calibration clips per corpus are evaluated on a GTX 1650 Ti in float32, at batch size one with two CPU

threads and one GPU worker. One, three, and five requested intervals are deduplicated when short clips yield identical queries. Development-selected systems and a separate matched pure-CTC control use the exact same clips and query schedules. In the matched control, every arm uses beam width 40 and token-pruning limit 64 without decoder recalibration. Three fresh sessions are run per scenario, with arm order rotated by clip and repetition. We first take within-clip medians, then summarize their median and 95th percentile across clips. Recognition and required allocation are timed with the model resident; input loading is measured separately. The experiment also measures repeat lookup from populated caches. It does not measure raw-video face tracking, live acquisition delay, or human editing time.

4.4. Uncertainty and comparison families

The eight principal contrasts are projection versus crop-first and projection versus contextual direct in each corpus. We use paired source-cluster bootstrap resampling with 10,000 draws. Bonferroni adjustment gives 99.375% individual percentile intervals for this family. The interval construction was chosen at analysis time for the fixed comparison family; it has approximate bootstrap coverage. Other intervals describe explanatory contrasts at pointwise 95% coverage. Limited source counts, especially in Spanish, constrain the precision and generality of those comparisons.

An explanatory analysis recomputes matched greedy and pure-CTC intervals from the frozen predictions and partitions each cohort by actual outside context: none, left only, right only, or both. These post-hoc explanatory comparisons use the same 10,000-draw paired-source procedure with pointwise 95% intervals; they do not alter the primary comparison family or decoder

Table 1: Requested-interval WER (%). Each arm uses its development-selected decoder. Targets are at most two seconds with up to 0.64 s of context per side. Source clusters are recording identifiers, not verified speakers.

Corpus	Clips	Sources	Crop A	Direct B	Projection C
VLR	11,306	61	58.602	50.579	50.434
LRS2	1,243	354	24.099	23.609	21.733
Spanish	934	12	85.451	81.527	80.170
ViCocktail	1,167	20	51.767	49.179	48.193

selection. Geometry is read from the recorded input and target coordinates.

5. Results

5.1. Context improves requested-interval recognition

Projection improves over crop-first recognition in all four principal comparisons (Table 1). The reduction ranges from 2.37 to 8.17 percentage points, and all four family-adjusted intervals favor projection. The result concerns the same requested target within each corpus: it is not a comparison between a short target transcript and a longer full-clip transcript.

Contextual direct decoding captures much of the benefit. For VLR, moving from crop-first to contextual direct changes WER from 58.602% to 50.579%; projection provides a further 0.145-point reduction. The additional selected-system reductions are 1.876 points on LRS2, 1.357 on Spanish, and 0.986 on ViCocktail. The adjusted Spanish C–B interval includes zero; the other three adjusted intervals favor C.

Matched decoders give a more specific interpretation (Table 2). Un-

Table 2: Matched-decoder requested-interval WER (%). Negative C–B favors projection. Brackets give explanatory pointwise 95% paired-source bootstrap intervals from 10,000 draws; the primary adjusted family is reported separately.

Corpus	Decoder	A	B	C	C–B [95% CI] (pp)
VLR	Greedy	58.602	50.579	50.434	-0.145 [-0.181, -0.110]
VLR	Pure CTC	58.971	50.862	50.691	-0.172 [-0.214, -0.129]
LRS2	Greedy	30.527	28.749	28.471	-0.277 [-0.463, -0.126]
LRS2	Pure CTC	30.250	28.536	28.226	-0.310 [-0.500, -0.155]
Spanish	Greedy	83.181	81.824	80.870	-0.954 [-1.390, -0.672]
Spanish	Pure CTC	83.203	81.718	80.764	-0.954 [-1.429, -0.670]
ViCocktail	Greedy	55.540	52.643	52.524	-0.119 [-0.213, -0.011]
ViCocktail	Pure CTC	55.520	52.623	52.464	-0.159 [-0.249, -0.061]

der pure CTC, the C–B improvements are 0.172, 0.310, 0.954, and 0.159 points, with explanatory paired-source 95% intervals favoring C in all four settings. Thus projection contributes beyond contextual state cropping at the principal operating point, while contextual encoding accounts for the larger improvement. The selected-system comparisons add the effect of each arm's development-selected decoder. Spanish illustrates the relevance of both views: crop-first greedy and pure CTC outperform its selected hybrid on assessment.

Decoder sensitivity.. Projection does not outperform contextual direct at every decoder setting. On the separate VLR development screen, pure CTC with zero blank bias gives 60.085% for C versus 59.659% for B: a C–B difference of +0.426 points with pointwise 95% interval [+0.105, +0.802]. VLR greedy decoding also favors B by point estimate at all three tested blank biases. On

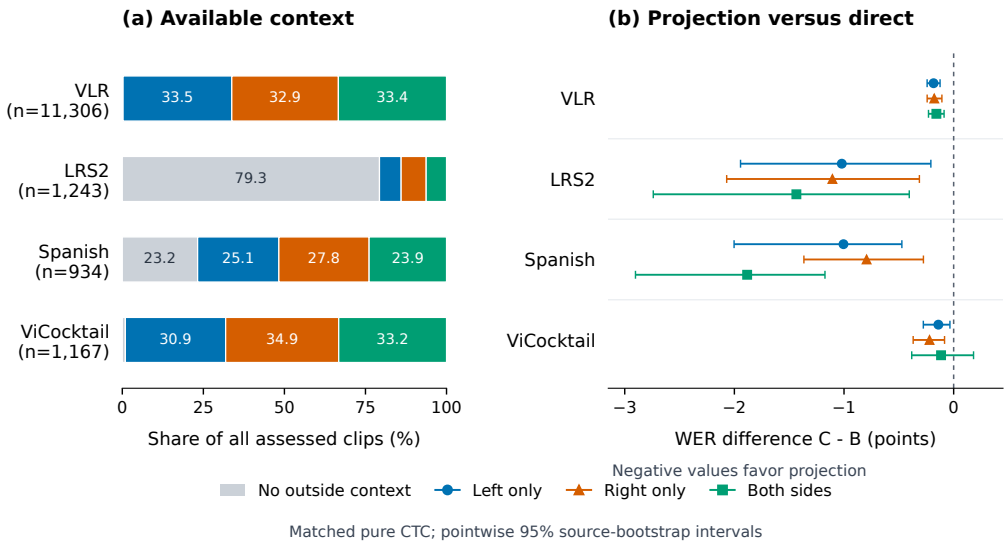


Figure 4: Post-hoc analysis of the frozen principal assessment, using the same pure-CTC decoder for A/B/C. (a) All clips partitioned by actual context outside the target. (b) C–B WER differences within the three groups with outside context; negative values favor projection. Bars are pointwise 95% paired-source bootstrap intervals, without multiplicity adjustment. No-context predictions are identical across A/B/C. Groups differ in request position and clip duration, so their differences do not isolate causal left/right effects. Full counts and WERs appear in Table A.9.

ViCocktail, a hybrid length bonus of +1 gives 45.302% for C versus 44.214% for B (+1.088 points, interval [-0.340, +2.739]); this comparison retains the respective CTC weights of 0.1 and 0.5. These development results show that the readout ranking depends on decoder configuration. They do not retune or replace the frozen assessment results (Appendix Appendix G).

5.2. Actual context availability explains aggregate readout effects

A post-hoc stratification separates requests by actual outside context (Figure 4; Table A.9). With matched decoders, A/B/C predictions are identical when the request covers the available video. This occurs on 986 of 1,243 LRS2 clips (79.3%), compared with 27/11,306 VLR, 217/934 Spanish, and 11/1,167 ViCocktail clips. The LRS2 aggregate thus mixes many identity cases with requests that permit contextual readout.

For the LRS2 groups with left, right, and both-sided context (83, 96, and 78 clips), the pure-CTC C–B reductions are 1.020, 1.106, and 1.434 points; each corresponding pointwise interval favors C. The reductions across these three groups range from 0.158 to 0.183 points on VLR and from 0.794 to 1.883 on Spanish. ViCocktail has smaller reductions of 0.115–0.220 points, and its both-context interval includes zero. The complete principal endpoint retains all clips, including those without outside context. This explanatory analysis uses pointwise 95% intervals on previously evaluated data; it is not an independent confirmation or a change to the primary comparison family. Directional effects of context are tested separately by the interventions below.

5.3. Visual alignment contributes to output selection

Table 3 compares allocation rules applied to the same contextual prediction. Aligned projection has lower error than both uniform rules in each principal setting. This comparison cannot be explained by a different generated transcript: the alternatives share predicted tokens and differ only in their assignment to the requested interval.

Unrestricted full-clip lookup has the lowest target error in these settings. It provides the accuracy reference when the application can process the entire

Table 3: Development-selected requested-interval WER (%). The uniform controls allocate the same decoded sequence as C. Full-clip lookup processes unrestricted input and returns the same requested target; it is a different input-budget baseline.

Corpus	Aligned C	Uniform words	Uniform tokens	Full-clip lookup
VLR	50.434	56.168	56.066	42.720
LRS2	21.733	22.842	23.071	21.553
Spanish	80.170	82.821	82.694	78.770
ViCocktail	48.193	52.125	52.364	45.495

video. The bounded alternatives answer the same request using a prescribed observation interval, exposing the accuracy cost of that input constraint.

5.4. Real context supplies more than additional frames

On clips with outside context, real video improves over repeated-boundary padding under both selected systems and matched pure CTC (Table 4). The target pixels and input length are unchanged in this contrast. Thus the benefit is not explained solely by a longer tensor or a different target crop. The complete intervention table in Appendix Appendix D also compares single-sided and reversed context. Both real margins outperform either margin alone in those settings, while the preferred individual side varies across corpus/model conditions.

These interventions establish the value of real neighboring content over the tested alternatives. They affect motion continuity together with speech information, so the contrast concerns visual context as a whole.

Table 4: Real context versus repeated-boundary padding, restricted to clips with available outside context. Both conditions preserve target pixels and input length. Intervals are explanatory pointwise 95% source-bootstrap intervals.

Corpus	Decoder regime	Clips	Real–repeat difference (pp)
VLR	Selected	254	-4.219 [-5.905, -2.605]
VLR	Pure CTC	254	-4.430 [-6.026, -2.913]
LRS2	Selected	48	-7.767 [-14.286, -2.222]
LRS2	Pure CTC	48	-7.443 [-12.333, -2.857]
Spanish	Selected	202	-9.635 [-11.979, -7.156]
Spanish	Pure CTC	202	-3.043 [-4.700, -1.230]
ViCocktail	Selected	254	-9.286 [-11.456, -7.326]
ViCocktail	Pure CTC	254	-6.066 [-8.193, -4.103]

5.5. Duration changes the opportunity to use context

The one-second LRS2 target provides a substantial context opportunity (Figure 5). Selected projection decreases from 47.48% without context to 32.151% at a 0.32-second per-side allowance and 29.98% at 0.64 seconds. At the latter budget, contextual direct is 35.049%. Under matched pure CTC, direct and projection are 37.974% and 36.566%, a 1.408-point reduction with pointwise source-bootstrap interval [1.023, 1.819] when expressed as an improvement.

The opportunity shrinks as the target covers more of these short clips. Outside context exists for 888 of 1,024 clips at a one-second target, 207 at two seconds, and only 53 at three seconds. With a one-second target and a nominal 0.64-second per-side budget, mean actual input is 1.526 seconds; mean total added context is 0.541 seconds.

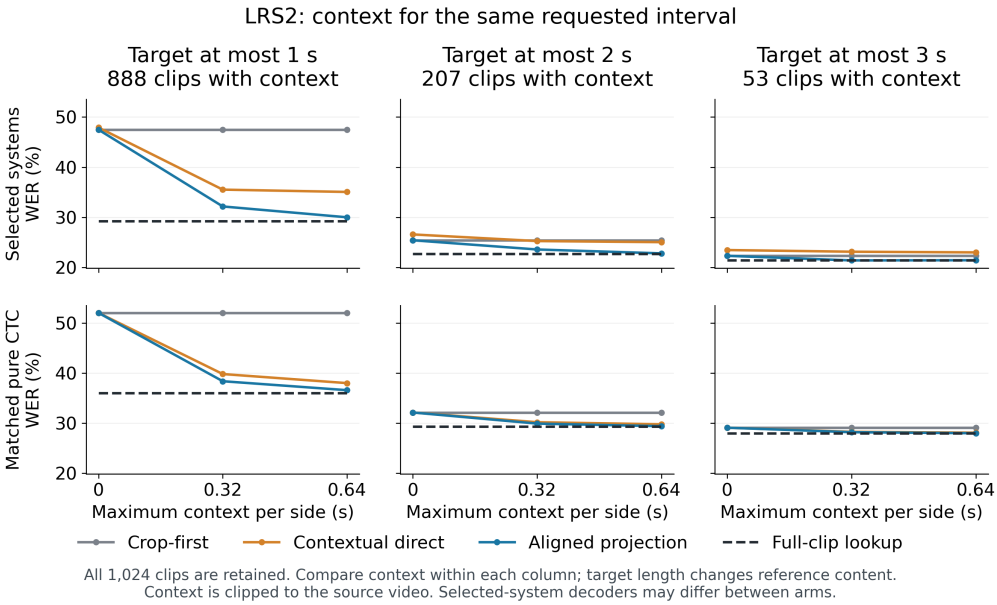


Figure 5: LRS2 duration experiment on 1,024 clips. Compare context budgets within a column: the target text is fixed there. Columns change the requested content and are not competing estimates of the same transcript. Both development-selected systems and matched pure CTC are shown.

The complete VLR grid provides a different duration distribution (Appendix Appendix C). Outside context exists for 1,022 clips at one- and two-second targets, and 1,012 at three seconds. With greedy decoding, one-second crop-first WER is 82.659%, contextual direct with the larger budget is 57.018%, and projection is 56.490%. Its 0.529-point advantage over direct has pointwise interval [0.222, 0.847] when expressed as a reduction. Pure CTC gives a corresponding 0.608-point reduction [0.230, 0.996]. At two- and three-second targets, the incremental greedy C–B differences are -0.053 and -0.044 points, with intervals including zero. Duration therefore affects both the availability of outside evidence and the benefit of retaining context through

decoding. Context budgets are compared within a fixed target; different target lengths request different text.

The Spanish and ViCocktail grids extend this comparison to different corpus/model conditions (Appendix Appendix C). At the 0.64-second allowance, selected Spanish projection changes from 105.737% to 87.330% for the one-second target, and from 89.499% to 79.991% for two seconds. Under matched pure CTC, its improvements over contextual direct are 2.270, 1.558, and 0.830 points for one-, two-, and three-second targets; the corresponding pointwise intervals all favor projection. ViCocktail shows a different readout profile. Selected projection improves from 70.109% to 59.447% for one-second requests, but the matched pure-CTC C–B difference is +0.067 points with interval [-0.153, +0.308]. At two and three seconds, the matched reductions are 0.133 and 0.102 points. A sizable context gain can therefore coexist with little additional projection gain.

Increasing context beyond 0.32 seconds also has a corpus-dependent return. On one-second targets with outside context, the matched-CTC change from 0.32 to 0.64 seconds per side is -3.574 points on VLR, -2.064 on LRS2, -0.585 on Spanish, and -2.345 on ViCocktail. The Spanish interval includes zero, whereas the other three favor the larger budget. These paired results support evaluating context budgets for the intended request and recognizer; they do not establish an adaptive policy or a universal optimal budget.

5.6. Whole-transcript recognition favors full input

Projected sliding can improve direct sliding, but full-clip recognition remains stronger on the same complete transcripts (Table 5). Joining contextual target CTC scores and decoding once removes some concatenation errors

Table 5: Whole-transcript WER (%) under matched pure CTC, 512 clips per corpus. The shared sliding schedule partitions each clip into balanced targets of at most two seconds with 0.64 s context per side. Full-clip output uses the same reference transcript. Intervals are pointwise source-bootstrap intervals.

Corpus	Direct sliding	Projected sliding	Full clip	C–F (pp)
VLR	48.667	48.452	42.877	+5.575 [+4.641, +6.491]
LRS2	28.936	28.186	27.286	+0.900 [+0.158, +1.631]
Spanish	78.866	77.353	74.558	+2.795 [+1.622, +4.239]
ViCocktail	47.723	47.691	43.756	+3.935 [+3.066, +4.743]

without closing that gap. At the two-second schedule, this control removes 25, 0, 2, and 10 errors relative to projected pure-CTC sliding on VLR, LRS2, Spanish, and ViCocktail.

The error balance varies across corpora. Under matched CTC, projected sliding relative to full-clip decoding adds 431 substitutions and 589 insertions on VLR while reducing deletions by 451; Spanish instead adds 220 deletions and 20 substitutions while reducing insertions by 33. These different error balances are compatible with disruption of recognition and repeated local decisions, extending beyond duplicated text at window seams.

5.7. Query workload determines the cost advantage

Holding the decoder fixed reveals a consistent workload distinction (Table 6). For one request, projection is faster than fresh full-clip lookup on 28/32 VLR clips, 22/32 LRS2 clips, 29/32 Spanish clips, and 28/32 ViCocktail clips. Their paired median C–F time differences are −0.612, −0.027, −0.378, and −0.083 seconds. For five requested intervals, the paired differences be-

Table 6: Matched pure-CTC response cost on 32 stratified calibration clips per corpus. Values are across-clip medians after reducing three fresh sessions to one median per clip. All arms use beam 40 with token pruning 64 on the same GTX 1650 Ti. The final column counts paired clips on which C is faster than F. Short clips can deduplicate requests; these descriptive timings do not measure assessment accuracy.

Corpus	Queries	A (s)	B (s)	C (s)	F (s)	C < F
VLR	1	0.390	0.432	0.623	1.221	28/32
VLR	3	1.159	1.269	1.658	1.034	2/32
VLR	5	1.954	2.128	2.865	1.093	0/32
LRS2	1	0.338	0.362	0.441	0.455	22/32
LRS2	3	1.026	0.979	0.459	0.431	6/32
LRS2	5	1.681	1.599	0.561	0.462	6/32
Spanish	1	0.378	0.419	0.589	0.987	29/32
Spanish	3	1.149	1.227	1.584	0.989	9/32
Spanish	5	1.903	2.037	2.785	1.014	0/32
ViCocktail	1	0.350	0.383	0.552	0.677	28/32
ViCocktail	3	1.033	1.132	1.458	0.741	10/32
ViCocktail	5	1.734	1.909	2.563	0.708	5/32

come +1.698, +0.105, +1.824, and +0.938 seconds. Full-clip recognition pays its initial cost once and reuses the resulting transcript and temporal index. Marginal medians in the table and medians of paired differences describe different summaries; they need not differ by the same amount.

Projection also has a cost relative to contextual direct decoding. The paired median C–B difference is positive in all twelve corpus/workload cells, ranging from 0.017 to 0.754 seconds. Encoding frame counts are identical

between B and C in every paired clip. C generally decodes a longer sequence and adds alignment, but distinct targets can sometimes share a contextual prediction. In LRS2, this reuse reduces C's decode count relative to B on seven clips with three requests and eight with five; C is faster on each of those clips. This is the cache behavior described in Section 3.5, not an additional accuracy gain.

The supplement reports selected-system costs, matched tail latency, and memory. Together with the paired accuracy results, these measurements distinguish the cost of a bounded request from the benefit of amortizing full-clip recognition over a session.

6. Discussion

6.1. Separating recognition support from the requested answer

The central result is that an output boundary need not be an input boundary. Keeping neighboring visual evidence improves recognition of a fixed request, and the real-versus-padding intervention shows that this evidence supplies more than additional frames. The interval formulation makes that distinction usable with existing recognizers: context is admitted at encoding, while output scope can be enforced at the state or transcript level.

The matched comparisons locate the value of projection more precisely. Contextual direct decoding captures most of the benefit of admitting context. Projection adds a smaller improvement at the principal operating point in all four corpora, with more substantial effects in several short-target conditions. The LRS2 availability analysis shows why an aggregate over short clips can

understate this opportunity: most two-second requests already cover the whole clip. Together with the decoder screen, the results support treating readout as a design decision conditioned on the recognizer and request, rather than using one output rule for every operating point.

6.2. Practical use and computational reuse

The intended application is passage-level visual transcription: an editor or viewer selects a time span and receives the words assigned to it, with a visible allowance for surrounding video. This is useful when visual input is the available speech signal and the full transcript has not already been computed. The present evidence measures recognition and computational cost; a user study would be needed to quantify editing-time or accessibility benefits.

Two workload regimes follow from the measurements. An isolated request can benefit from processing a bounded passage; repeated requests can amortize a full-clip transcript and temporal index. Full-clip recognition also remains stronger for complete transcripts. Within bounded processing, direct readout offers the lower measured median cost, while projection trades additional decoding and alignment for its accuracy benefit. These operating distinctions give applications a concrete basis for selecting an inference strategy.

6.3. Scope of the evidence

Word timing is automatic. The tested reference perturbations preserve the favorable C–A and C–B point-difference signs, including matched decoders, but fine allocation is more sensitive: a +160 ms shift makes uniform-word allocation better than aligned projection by 1.041 points on VLR and 0.245 on LRS2. Independent review of boundary words would clarify this finer

comparison. The four corpus/model settings also differ in duration, training history, and source diversity; Spanish has 12 recording sources and high absolute error. The assessment material was available in earlier research. Results therefore support the stated within-corpus comparisons, with prospective source-held replication and evaluation outside the present recognizer family as valuable extensions. The cost measurements characterize a stratified local-GPU sample.

6.4. Toward request-dependent computation

The next methodological question is whether observable request geometry and predictive uncertainty can select context or readout more efficiently than a fixed strategy. Such a policy should be developed on separate sources and evaluated against fixed context budgets, contextual direct, projection, and cached full-clip lookup under compatible input permissions. Its feature extraction and routing costs must be included. The current duration and workload results provide the empirical basis for this question; they are not an evaluation of an adaptive policy.

7. Conclusion

Context-aware interval transcription separates the evidence used to recognize speech from the span of text requested by the user. Prediction-aligned projection implements this separation using a pretrained visual recognizer and its native CTC scores. Four-corpus experiments show improved requested-interval recognition over input cropping, while matched decoders distinguish contextual encoding from the incremental effect of transcript projection. Context availability and computational reuse explain the practical operating

conditions. This formulation extends pretrained VSR to selective passage requests and provides an empirical basis for balancing context, readout, and work across queries.

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Table A.7: Geometry and source concentration of the principal E1 cohorts. Mean and median describe prepared clip duration. No context means the at-most-two-second target covers the clip. Added is the mean total observed context across both sides under the frozen placement rule. Max. source is the largest recording source's share of target reference units.

Corpus	Mean (s)	Median [IQR] (s)	No context (%)	Added (s)	Max. source (%)
VLR	5.782	5.64 [4.96, 6.44]	0.2	0.851	4.8
LRS2	1.642	1.44 [1.16, 1.92]	79.3	0.099	3.2
Spanish	5.913	4.30 [2.16, 7.79]	23.2	0.596	24.9
ViCocktail	5.830	5.16 [4.24, 7.00]	0.9	0.843	13.4

Table A.8: Ranges after omitting one recording source at a time from the principal selected-system comparison. These are descriptive source-influence ranges, not confidence intervals. The primary source-bootstrap uncertainty remains in Table B.10.

Corpus	C–A range (pp)	C–B range (pp)
VLR	[-8.278, -7.990]	[-0.152, -0.139]
LRS2	[-2.421, -2.226]	[-1.962, -1.784]
Spanish	[-5.604, -4.969]	[-1.751, -1.111]
ViCocktail	[-3.919, -3.370]	[-1.210, -0.833]

Appendix A. Corpus geometry and source weighting

The duration statistics in Table A.7 describe the exact principal assessment cohorts, measured from prepared frames at 25 fps. They are not estimates of the complete source datasets. The interval in the median column gives the 25th and 75th percentiles. For the two-second request, 79.3% of LRS2 clips already fit inside the target, compared with 0.2% on VLR, 23.2% on Spanish, and 0.9% on ViCocktail. The principal hash-based start, center

and end placements further limit which margin is available. Mean observed context is therefore smaller than the maximum 1.28 seconds across both sides, even in corpora with substantially longer clips.

Recording sources also have different weights. The largest Spanish source contributes 24.9% of the target reference units, compared with 4.8% for VLR, 3.2% for LRS2 and 13.4% for ViCocktail. Both principal point differences remain favorable to projection after omitting any single source (Table A.8). These descriptive ranges address the influence of one recording; they do not imply significance or replace the source-bootstrap intervals. In particular, the adjusted Spanish C–B interval still includes zero. Duration, recording composition and checkpoint history should consequently accompany any interpretation across corpora.

Appendix A.1. Recognition by actual context availability

Table A.9 separates the principal assessment by whether the recorded input adds left, right, both, or no context to the target. It uses matched pure CTC throughout. With no outside context, A/B/C return identical predictions on all 27 VLR, 986 LRS2, 217 Spanish, and 11 ViCocktail clips. The whole-cohort difference therefore combines an identity subset with clips on which contextual inference can change the result.

The three LRS2 outside-context groups show C–B reductions of 1.020–1.434 points. VLR's corresponding reductions are 0.158–0.183 points, and Spanish's are 0.794–1.883 points. ViCocktail shows smaller differences, with the interval for its both-context group including zero. This decomposition complements the duration experiment without excluding short clips from the principal endpoint. The groups differ in target position and clip duration;

comparisons across groups are descriptive rather than controlled left-versus-right interventions.

Appendix B. Primary comparison uncertainty

Table B.10 reports the complete primary family. Source resampling preserves paired arm predictions within each sampled recording. It addresses within-source dependence more directly than treating every clip as independent, but it cannot account for unobserved dependence between recordings or repair prior exposure to the assessment corpus. In particular, the Spanish C–B interval includes zero after the family adjustment.

Appendix C. Complete duration grids and actual context

Table C.11: Complete duration grids: development-selected WER (%). Cohort sizes are VLR: 1,024; LRS2: 1,024; Spanish: 934; ViCocktail: 1,024. Target length and per-side context are maxima clipped to available video. Compare context within a fixed target; target length changes reference content. Decoders: VLR A/B/C: greedy/greedy/greedy; LRS2 A/B/C: hybrid10/hybrid50/hybrid10; Spanish A/B/C: hybrid10/hybrid50/hybrid10; ViCocktail A/B/C: hybrid10/hybrid50/hybrid10.

Corpus	Target (s)	Context (s)	Units	A	B	C
VLR	1	0	3783	82.659	82.659	82.659
VLR	1	0.32	3783	82.659	60.455	60.005
VLR	1	0.64	3783	82.659	57.018	56.490

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Corpus	Target (s)	Context (s)	Units	A	B	C
VLR	2	0	7608	59.727	59.727	59.727
VLR	2	0.32	7608	59.727	49.882	49.724
VLR	2	0.64	7608	59.727	48.686	48.633
VLR	3	0	11449	50.808	50.808	50.808
VLR	3	0.32	11449	50.808	46.161	46.161
VLR	3	0.64	11449	50.808	45.061	45.017
LRS2	1	0	3692	47.481	47.914	47.481
LRS2	1	0.32	3692	47.481	35.509	32.151
LRS2	1	0.64	3692	47.481	35.049	29.984
LRS2	2	0	5052	25.396	26.564	25.396
LRS2	2	0.32	5052	25.396	25.238	23.555
LRS2	2	0.64	5052	25.396	25.020	22.783
LRS2	3	0	5353	22.287	23.445	22.287
LRS2	3	0.32	5353	22.287	23.109	21.371
LRS2	3	0.64	5353	22.287	22.978	21.409
Spanish	1	0	2423	105.737	102.889	105.737
Spanish	1	0.32	2423	105.737	92.035	92.778
Spanish	1	0.64	2423	105.737	89.889	87.330
Spanish	2	0	4428	89.499	89.273	89.499
Spanish	2	0.32	4428	89.499	86.134	83.424
Spanish	2	0.64	4428	89.499	83.853	79.991
Spanish	3	0	6143	79.993	81.605	79.993
Spanish	3	0.32	6143	79.993	80.905	78.187
Spanish	3	0.64	6143	79.993	81.345	77.601
ViCocktail	1	0	4483	70.109	69.708	70.109
ViCocktail	1	0.32	4483	70.109	62.614	61.923
ViCocktail	1	0.64	4483	70.109	60.228	59.447
ViCocktail	2	0	9019	54.928	54.862	54.928

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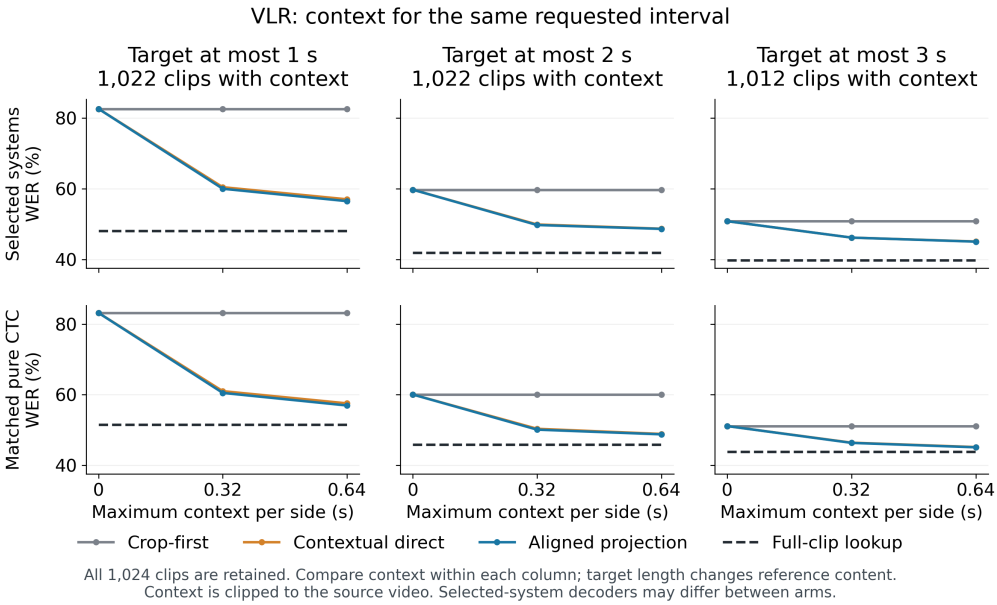


Figure C.6: VLR duration grid on the complete 1,024-clip cohort. Contextual direct and projection are close under the same greedy or pure-CTC decoder. The selected full-clip lookup uses a hybrid decoder, so its separation from the selected greedy systems includes a decoder difference.

Corpus	Target (s)	Context (s)	Units	A	B	C
ViCocktail	2	0.32	9019	54.928	51.591	50.815
ViCocktail	2	0.64	9019	54.928	50.194	48.841
ViCocktail	3	0	13733	49.290	49.712	49.290
ViCocktail	3	0.32	13733	49.290	46.843	46.137
ViCocktail	3	0.64	13733	49.290	46.232	45.453

The nominal target and context lengths are converted to 25, 50, or 75 target frames and 0, 8, or 16 context frames per side. All duration cells use centered targets, while the principal two-second assessment includes

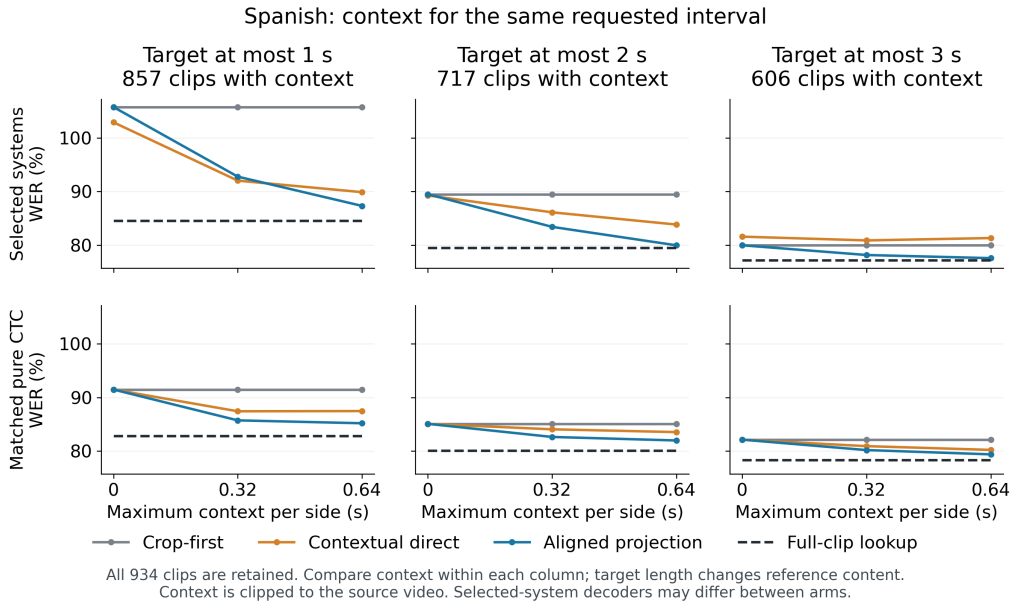


Figure C.7: Spanish duration grid on all 934 eligible clips. Native hybrid decoding and pure CTC have different error levels, but both support examining context within a fixed target. The same decoder is retained across context budgets within each arm. Absolute error remains high on this evaluation cohort.

deterministic start, center, and end placements. Consequently, the duration grid is not an exact repeat of the principal table even where the nominal lengths coincide.

At zero context, crop-first and projected output agree when the decoder is the same. This is an identity control, not an additional recognition gain. The full-clip target baseline is also invariant to the requested context budget at a fixed target. Table C.12 makes clipping at video boundaries explicit. Clips without outside context remain in every whole-cohort comparison.

Table C.13 keeps the incremental C–B comparisons for every target and corpus. The Spanish matched-CTC reductions span 0.830–2.270 points,

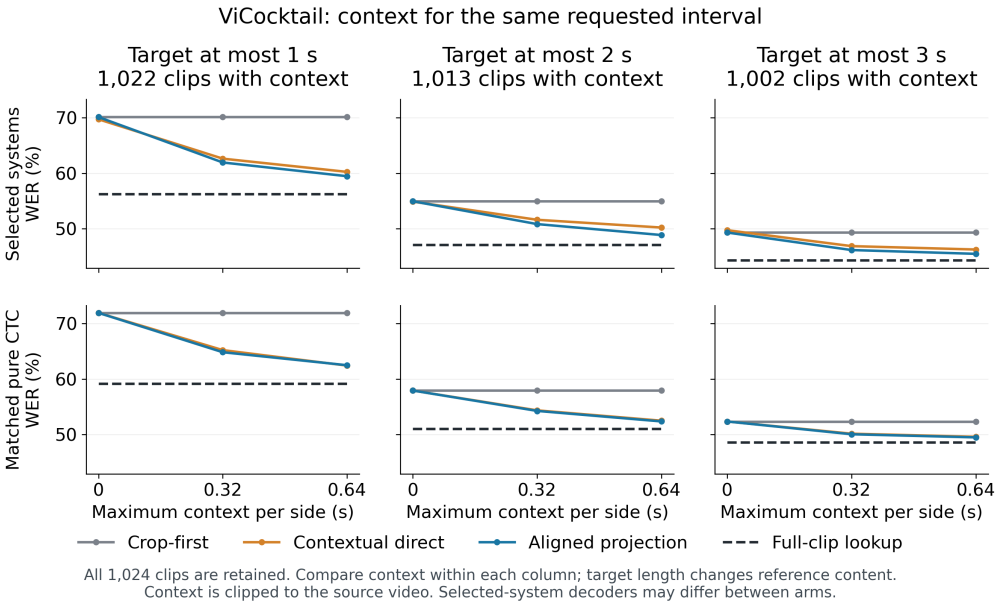


Figure C.8: ViCocktail duration grid on 1,024 clips. Context improves target recognition, while matched pure-CTC direct and projected readout remain close. The additional selected-system projection gain includes the difference between development-selected hybrid weights.

whereas the ViCocktail matched differences stay within 0.133 points of zero. The two-second Spanish C–B comparison is stronger in this centered duration grid than in the principal placement mixture. Different placement and cohort composition make these complementary comparisons; the duration result does not replace the principal family-adjusted interval. Spanish one-second selected WER can exceed 100% because insertions contribute to the numerator. The underlying edit counts, rather than a capped percentage, are retained in every comparison.

Appendix D. Complete context interventions

Table D.14 keeps all five context conditions on the same outside-context subset in each corpus. Both real margins give lower projected WER than either margin alone in every displayed setting. The relative value of the two individual margins is not constant: selected VLR projection gives 51.741% with the left margin and 57.806% with the right, whereas ViCocktail gives 54.876% and 54.130%, respectively. Removing a margin also shortens the encoder input and changes the native output cap, so these differences are operating-condition comparisons rather than isolated measures of directional linguistic information. They do not establish a language-specific rule.

Reversing each margin separately preserves its frames and length but changes their temporal order and continuity with the target. Real context improves the selected projected output by 5.327 points on VLR, 8.114 on Spanish and 10.079 on ViCocktail, with pointwise intervals excluding zero. LRS2 gives a 4.207-point improvement, but its interval includes zero on the smaller 48-clip subset. The matched pure-CTC contrasts have the same distinction. Thus repeated-boundary and reversed context probe different properties; neither intervention alone identifies the source of the recognition benefit.

For the real-versus-repeat contrast, omitting any one recording source keeps the point difference favorable in all four corpora under both reported decoder regimes. In Spanish, the selected-system difference ranges from -10.284 to -8.970 points across these omissions, and the matched-CTC difference ranges from -3.654 to -2.632. This check reduces concern that one source alone determines the observed effect; it does not replace evaluation

on additional sources.

Appendix E. Query-cost measurements

The query schedules for one, three, and five requested intervals are separate scenarios, not nested prefixes of one common schedule. Median cost is therefore not required to increase monotonically between scenarios, especially after deduplication on short clips. Full-clip lookup caches a complete recognition and alignment result; projected inference recognizes the bounded inputs required by the distinct requests.

Table E.16 complements the matched-decoder median costs with the 95th percentile and peak allocated GPU memory. These percentiles describe the stratified sample of per-clip medians; they do not estimate a deployment-population latency distribution. The accompanying numerical results retain duration strata, actual distinct query counts, first and later responses, cached responses, and computation counts. Duration strata are at most 2 seconds, (2,4], (4,8], and above 8 seconds; empty strata are omitted.

The timing endpoints include native normalization, device transfer, decoding, and the alignment needed by the output procedure. Prepared-input reading and decompression are measured separately. GPU memory is peak allocated tensor memory rather than total device usage. The calibration timing experiment adds no assessment WER measurements and cannot establish a universal break-even number of queries.

Appendix F. Sensitivity to reference timing

The seven midpoint conditions shift the reference word times by -160 ms, -80 ms, -40 ms, 0 ms, +40 ms, +80 ms, and +160 ms while leaving target geometry and predictions fixed. The eighth condition uses any positive temporal overlap at zero offset. Whole-clip targets retain their original full transcript. These changes can alter reference membership and the denominator, so each condition compares arms only against its own shared reference.

Table F.17 reports the full range of differences, rather than selecting a favorable clock shift. C–A and C–B keep negative point differences throughout the tested conditions. C–uniform is sensitive enough to reverse on VLR and LRS2. No clock offset or ownership convention is selected from these assessment results.

Appendix G. Sensitivity to decoder settings

A separate development screen covers 256 VLR, 256 LRS2, 143 Spanish, and 256 ViCocktail clips. These clips are disjoint from the calibration clips and from assessment sources. With the two-second target and the 0.64-second context allowance fixed, greedy and pure-CTC decoding use blank-logit biases of -0.5, 0, and $+0.5$. Native hybrid decoding uses length bonuses of -1, 0, and $+1$, retaining its previously chosen CTC/attention weight. This screen examines stability and does not select a replacement decoder.

Projection improves on crop-first at all nine settings in every corpus. Its smaller advantage over contextual direct is less consistent. For VLR, the pure-CTC C–B difference ranges from -0.213 to $+0.426$ points; the zero-bias setting gives 60.085% for C and 59.659% for B, corresponding to eight

additional errors. For ViCocktail, pure-CTC differences remain small and favorable (-0.198 to -0.049), whereas the hybrid difference ranges from -0.495 to $+1.088$ as the length bonus changes. These findings reinforce the distinction between the broad benefit of context and the decoder-dependent value of projection. The development screen is not pooled with the principal assessment or used to replace its frozen predictions.

Appendix H. Implementation scope

Visual-only inference retains native preprocessing, tokenizers, blank and end-of-sequence identities, and vocabularies (2,002 classes for VLR and ViCocktail, 5,049 for English, and 1,006 for Spanish). Audio-derived reference times are used only for evaluation. Word grouping preserves native boundary symbols and decodes byte tokens before Unicode normalization. Infeasible alignment falls back to uniform allocation; this fallback was unused in the reported principal, context-intervention, and duration cohorts. Whole-input targets return the native transcript unchanged.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The accompanying reproducibility package contains numerical results, source-level edit counts, evaluation settings and checkpoint identities. VLR is

available from its versioned Zenodo record (Nguyen Quoc et al., 2026). LRS2 is obtained under the Oxford–BBC data-sharing agreement; MuAViC/mTEDx and ViCocktail are obtained from their original releases (Afouras et al., 2022; Anwar et al., 2023; Nguyen et al., 2025). Original media and third-party checkpoints retain their respective access terms. The derived numerical tables in the companion release are prepared for CC BY 4.0; this does not relicense any source recording, restricted metadata, or checkpoint. The package records the provenance of the VLR image examples separately.

Code availability

The accompanying package provides the interval geometry, CTC decoding, prediction alignment, word allocation and sliding controls, together with tests and scripts for reproducing the principal numerical comparisons. Model-specific code and asset requirements are documented separately from the lightweight analysis workflow. The reviewed source release is available in the research subtree of the public GitHub repository ntthienphuc/liptrace. The exact release commit is recorded in the accompanying handoff. Original study code is prepared for MIT licensing and the derived numerical tables for CC BY 4.0; the notices do not relicense source media, restricted metadata, checkpoints or third-party code. The existing VLR v2 Zenodo record is cited for the dataset; no separate software DOI has been created.

Table A.9: Post-hoc context-availability analysis of the frozen principal assessment, with the same pure-CTC decoder for A/B/C. WER is in percent; C–B differences are in percentage points with explanatory pointwise 95% paired-source intervals, without multiplicity adjustment. Every clip is retained. Availability groups follow the fixed request geometry and are descriptive.

Corpus	Context	Clips	A	B	C	C–B [95% CI]
VLR	None	27	89.195	89.195	89.195	+0.000 [+0.000, +0.000]
VLR	Left	3,791	59.678	47.809	47.626	-0.183 [-0.242, -0.124]
VLR	Right	3,717	55.839	55.078	54.901	-0.177 [-0.242, -0.107]
VLR	Both	3,771	61.056	48.816	48.658	-0.158 [-0.229, -0.087]
LRS2	None	986	29.836	29.836	29.836	+0.000 [+0.000, +0.000]
LRS2	Left	83	33.673	25.510	24.490	-1.020 [-1.944, -0.208]
LRS2	Right	96	27.014	23.381	22.275	-1.106 [-2.070, -0.312]
LRS2	Both	78	34.836	26.230	24.795	-1.434 [-2.740, -0.405]
Spanish	None	217	84.358	84.358	84.358	+0.000 [+0.000, +0.000]
Spanish	Left	234	81.993	78.727	77.722	-1.005 [-2.003, -0.472]
Spanish	Right	260	81.271	80.146	79.351	-0.794 [-1.366, -0.276]
Spanish	Both	223	86.188	84.933	83.049	-1.883 [-2.902, -1.174]
ViCocktail	None	11	59.649	59.649	59.649	+0.000 [+0.000, +0.000]
ViCocktail	Left	361	54.862	53.120	52.980	-0.139 [-0.277, -0.034]
ViCocktail	Right	407	52.642	50.605	50.385	-0.220 [-0.369, -0.083]
ViCocktail	Both	388	58.996	54.204	54.089	-0.115 [-0.382, +0.181]

Table B.10: Paired source-bootstrap intervals for the eight primary system comparisons. Negative differences favor projection. Bonferroni adjustment uses 99.375% individual percentile intervals from 10,000 draws; coverage is approximate.

Corpus	Contrast	Difference and family-adjusted interval (pp)
VLR	C–A	-8.168 [-9.086, -7.282]
VLR	C–B	-0.145 [-0.194, -0.097]
LRS2	C–A	-2.366 [-3.377, -1.449]
LRS2	C–B	-1.876 [-2.916, -0.852]
Spanish	C–A	-5.281 [-7.487, -3.903]
Spanish	C–B	-1.357 [-3.025, +0.020]
ViCocktail	C–A	-3.574 [-5.126, -2.001]
ViCocktail	C–B	-0.986 [-1.834, -0.183]

Table C.12: Actual input geometry at a maximum of 0.64 s context per side. Context available counts clips with any outside video; both full counts clips with two complete margins. Input and added context are per-clip means, with added context summed across both sides.

Corpus	Target (s)	Clips	Context available	Both full	Input (s)	Added (s)
VLR	1	1024	1022	1021	2.276	1.277
VLR	2	1024	1022	1011	3.268	1.271
VLR	3	1024	1012	910	4.219	1.228
LRS2	1	1024	888	150	1.526	0.541
LRS2	2	1024	207	42	1.604	0.127
LRS2	3	1024	53	13	1.626	0.035
Spanish	1	934	857	687	2.036	1.056
Spanish	2	934	717	580	2.712	0.886
Spanish	3	934	606	477	3.275	0.741
ViCocktail	1	1024	1022	1012	2.272	1.272
ViCocktail	2	1024	1013	1000	3.253	1.258
ViCocktail	3	1024	1002	753	4.141	1.162

Table C.13: Incremental projection effect relative to contextual direct at a maximum of 0.64 s context per side. Negative differences favor projection. Brackets are explanatory 95% source-cluster bootstrap intervals, not multiplicity-adjusted confirmatory tests. Each contrast compares the same target and cohort.

Corpus	Target (s)	Selected C–B (pp)	Pure CTC C–B (pp)
VLR	1	-0.529 [-0.847, -0.222]	-0.608 [-0.996, -0.230]
VLR	2	-0.053 [-0.221, +0.108]	-0.118 [-0.281, +0.027]
VLR	3	-0.044 [-0.104, +0.017]	-0.079 [-0.156, +0.000]
LRS2	1	-5.065 [-6.299, -3.849]	-1.408 [-1.819, -1.023]
LRS2	2	-2.237 [-3.136, -1.368]	-0.376 [-0.583, -0.191]
LRS2	3	-1.569 [-2.325, -0.821]	-0.112 [-0.205, -0.037]
Spanish	1	-2.559 [-4.405, -0.330]	-2.270 [-3.435, -1.430]
Spanish	2	-3.862 [-5.241, -2.146]	-1.558 [-2.205, -1.081]
Spanish	3	-3.744 [-5.054, -2.697]	-0.830 [-1.193, -0.579]
ViCocktail	1	-0.781 [-1.677, +0.035]	+0.067 [-0.153, +0.308]
ViCocktail	2	-1.353 [-2.066, -0.690]	-0.133 [-0.248, -0.012]
ViCocktail	3	-0.779 [-1.137, -0.320]	-0.102 [-0.194, -0.013]

Table D.14: Complete context interventions: requested-interval WER (%) on the same clips with outside context within each corpus (VLR 254, LRS2 48, Spanish 202, ViCocktail 254). Targets are centered and at most two seconds. Real, repeat and reverse preserve input length; left/right retain only that margin and therefore change length. Selected A/C use greedy for VLR and hybrid10 otherwise; pure CTC is matched throughout.

Corpus	Regime	Crop A	Real C	Left C	Right C	Repeat C	Reverse C
VLR	Selected	60.021	50.211	51.741	57.806	54.430	55.538
VLR	Pure CTC	60.549	49.947	51.688	57.753	54.378	55.643
LRS2	Selected	27.508	18.123	23.948	24.272	25.890	22.330
LRS2	Pure CTC	33.333	23.625	28.479	28.479	31.068	26.861
Spanish	Selected	84.686	70.690	76.775	81.947	80.325	78.803
Spanish	Pure CTC	80.832	75.963	77.688	79.615	79.006	79.716
ViCocktail	Selected	57.116	50.537	54.876	54.130	59.823	60.616
ViCocktail	Pure CTC	60.429	54.550	57.210	57.163	60.616	61.223

Table E.15: Median cumulative fresh-session response time on 32 stratified calibration clips per corpus. Three sessions are reduced to per-clip medians first. Repeated identical intervals on short clips are deduplicated. These are selected-system measurements on a GTX 1650 Ti, not matched-decoder speedups or assessment accuracy.

Corpus	Requested queries	A (s)	B (s)	C (s)	F (s)
VLR	1	0.037	0.055	0.061	4.250
VLR	3	0.113	0.155	0.167	4.305
VLR	5	0.190	0.266	0.286	4.244
LRS2	1	1.626	1.379	1.825	1.728
LRS2	3	4.387	3.902	2.026	1.867
LRS2	5	6.938	6.584	1.913	1.877
Spanish	1	1.870	1.623	3.384	6.212
Spanish	3	5.662	4.791	8.464	6.726
Spanish	5	9.256	7.872	14.321	6.513
ViCocktail	1	1.659	1.541	2.661	3.753
ViCocktail	3	4.755	4.465	6.883	3.740
ViCocktail	5	7.863	7.491	11.422	3.732

Table E.16: Tail response cost and GPU memory under matched pure CTC. The p95 is across 32 per-clip medians of cumulative fresh-session time; memory is the median per-clip peak allocated tensor memory, including the resident model. Prepared-input loading is separate. The sample is duration-stratified, so its p95 is not a deployment-population tail estimate.

Corpus	Queries	C p95 (s)	F p95 (s)	C peak (MiB)	F peak (MiB)
VLR	1	0.71	1.70	1104.78	1179.58
VLR	5	3.45	1.81	1105.45	1178.28
LRS2	1	0.63	0.82	1101.27	1101.27
LRS2	5	2.75	0.86	1101.27	1101.27
Spanish	1	0.66	2.28	1105.37	1182.09
Spanish	5	3.21	2.41	1106.04	1180.78
ViCocktail	1	0.63	2.03	1098.95	1118.62
ViCocktail	5	2.99	2.20	1099.07	1117.32

Table F.17: Ranges of point differences across seven midpoint clock settings and one overlap convention. These are sensitivity ranges, not confidence intervals. Negative differences favor C.

Corpus	Regime	C–A (pp)	C–B (pp)	C–Uniform (pp)
VLR	Selected	[-12.131, -2.431]	[-0.206, -0.011]	[-8.228, +1.041]
VLR	Pure CTC	[-12.237, -2.571]	[-0.205, -0.029]	[-8.217, +1.095]
LRS2	Selected	[-2.655, -1.766]	[-1.894, -1.728]	[-1.454, +0.245]
LRS2	Pure CTC	[-2.301, -1.161]	[-0.310, -0.165]	[-1.324, +0.213]
Spanish	Selected	[-6.583, -2.905]	[-1.543, -0.944]	[-3.700, -0.233]
Spanish	Pure CTC	[-3.379, -0.615]	[-0.954, -0.462]	[-2.962, -0.212]
ViCocktail	Selected	[-4.071, -2.312]	[-1.176, -0.385]	[-4.235, -2.262]
ViCocktail	Pure CTC	[-3.631, -1.940]	[-0.159, -0.051]	[-3.748, -2.071]